

The Foundations: Logic and Proofs

Chapter 1, Part I: Propositional Logic

Edited by Shih-Tsung Liang

Chapter Summary

- Propositional Logic (命題邏輯)
 - The Language of Propositions
 - Applications
 - Logical Equivalences
- Predicate Logic (述詞邏輯)
 - Predicates(述詞) and Quantifiers(量詞)
 - Logical Equivalences (等價)
 - Nested Quantifiers
- Proofs
 - Rules of Inference
 - Proof Methods
 - Proof Strategy

Propositional Logic (命題邏輯)

Section 1.1

Propositions

- A *proposition* (命題) is a declarative sentence (陳述句) that is either *true or false*.
- Examples of propositions:
 - a) The Moon is made of green cheese.
 - b) Toronto is the capital of Canada.
 - c) $1 + 0 = 1$
 - d) $0 + 0 = 2$
- Examples that are *not* propositions.
 - a) Sit down!
 - b) What time is it?
 - c) $x + 1 = 2$
 - d) $x + y = z$

Constructing Propositions

- Propositional Variables: $p, q, r, s, \dots$
- The proposition that is **always true** is denoted by **T** and the proposition that is **always false** is denoted by **F**.
- Compound Propositions: constructed from **logical connectives** (邏輯連接詞) and other propositions
 - Negation (否定) $\neg$
 - Conjunction (合取) $\wedge$
 - Disjunction (分取) $\vee$
 - Implication (蘊涵) $\rightarrow$
 - Biconditional (雙條件；若且唯若) $\leftrightarrow$

Compound Propositions: Negation

- The *negation* (否定) of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

- **Example:** If p denotes “The earth is round.”, then $\neg p$ denotes “**It is not the case** that the earth is round,” or more simply “The earth is not round.”

Conjunction (合取)

- The *conjunction* of propositions p and q is denoted by $p \wedge q$ and has this truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home and it is raining.”

Disjunction(分取)

- The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes “I am at home or it is raining.”

The Connective Or in English

- In English “or” has two distinct meanings.
 - “Inclusive Or”
 - “Students who have taken CS202 or Math120 may take this class”
 - This is the meaning of **disjunction**. For $p \vee q$ to be true, either one or both of p and q must be true.
 - “Exclusive Or” (互斥或)
 - “Soup or salad comes with this entrée”
 - This is the meaning of Exclusive Or (Xor).
 - The **exclusive or** of p and q , denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication(蘊涵)

- If p and q are propositions, then $p \rightarrow q$ is a *conditional statement* or *implication* which is read as “if p , then q ” and has this truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “If I am at home then it is raining.”
- In $p \rightarrow q$, p is the *hypothesis* (or *premise*) and q is the *conclusion* (or *consequence*).

Understanding Implication

- considering an obligation or contract. ($p \rightarrow q$)
 - “If **I am elected**, then **I will lower taxes.**”
- If the politician **is elected** and does **not lower taxes**, then the voters can say that he or she has broken the campaign pledge.
- This corresponds to the case where p is true and q is false.
- How about “If **you get 100% on the final (p)**, then **you will get an A(q) ?**”

Different Ways of Expressing $p \rightarrow q$

(Reading exercise)

- if p , then q
- if p , q
- q unless $\neg p$
- q if p
- q whenever p
- q follows from p
- p implies q
- p only if q
- q when p
- q provided that p
- p is sufficient(充分) for q
- q is necessary(必要) for p (i.e., 非 q 莫 p)
- a necessary condition for p is q
- a sufficient condition for q is p

Converse (逆命題), Contrapositive(逆否命題;對位命題), and Inverse(否命題; 反命題)

- From $p \rightarrow q$ we can form new conditional statements .
 - $q \rightarrow p$ is the **converse** of $p \rightarrow q$
 - $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$
 - $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It raining is a sufficient condition for my not going to town.”

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

Biconditional (若且唯若)

- If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as “ p **if and only if** q .” The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”

Expressing the Biconditional

- Some alternative ways “ p if and only if q ” is expressed in English:
 - p is **necessary and sufficient** for q
 - if p then q , and **conversely**
 - p **iff** q

Truth Tables For Compound Propositions

- Construction of a **truth table**:
- Rows
 - Need a row for **every possible combination** of values for the **atomic propositions**.
- Columns
 - Need a column for the compound proposition (usually at far right)
 - Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
 - This includes the atomic propositions

$$p \vee q \rightarrow \neg r$$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent (等價) Propositions

- Two propositions are **equivalent** if they always have the same truth value.
- Example:** Show using a truth table that the **implication** is equivalent to its **contrapositive**.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the **inverse** nor **converse** of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Precedence(優先順序) of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

If the intended meaning is $p \vee (q \rightarrow \neg r)$

then parentheses(括號) must be used.

Applications of Propositional Logic

Section 1.2

Translating English Sentences

- Steps to convert an English sentence to a statement in propositional logic
 1. Identify **atomic propositions** and represent using **propositional variables**.
 2. Determine appropriate **logical connectives**
- “If I go to Harry’s or to the country, I will not go shopping.”
 - p : I go to Harry’s
 - q : I go to the country.
 - r : I will go shopping.

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

One Solution: Let a , c , and f be propositions

a : “You can access the internet from campus,”

c : “You are a computer science major,” and

f : “You are a freshman.”

=>ans: $a \rightarrow (c \vee \neg f)$

System Specifications (規格)

- System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply cannot be sent **when** the file system is full”

Solution: One possible solution: Let p denote “The automated reply can be sent” and q denote “The file system is full.”

$$q \rightarrow \neg p$$

Consistent(一致的) System Specifications

Definition: A list of propositions is *consistent* if it is possible to assign truth values to the proposition variables so that each proposition is true.

Exercise: Are these specifications consistent?

- “The diagnostic message is stored in the buffer or it is retransmitted.”
- “The diagnostic message is not stored in the buffer.”
- “If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution:

Let p and q be propositions and

p : “The diagnostic message is stored in the buffer.”

q : “The diagnostic message is retransmitted”

The specification can be written as: $p \vee q$, $\neg p$, $p \rightarrow q$.

=> When p is false and q is true all three statements are true. So the specification is consistent.

What if “The diagnostic message is not retransmitted” is added. ?

Solution: Now we are adding $\neg q$ and there is no satisfying assignment. So the specification is not consistent.

Logic Puzzles



Raymond
Smullyan
(Born 1919)

- An island has two kinds of inhabitants, *knight*s(武士), who always tell the truth, and *knave*s(無頼), who always lie.
- You go to the island and meet A and B.
 - A says “B is a knight.”
 - B says “The two of us are of opposite types.”

Example: What are the types of A and B?

Solution: Let p and q be propositions and

P : “A is a knight”

q : “B is a knight”

$\Rightarrow \neg p$: “A is a knave” and

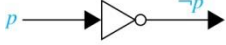
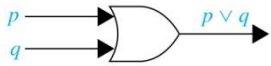
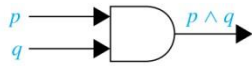
$\neg q$: “B is a knave”.

- If A is a knight, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- If A is a knave, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

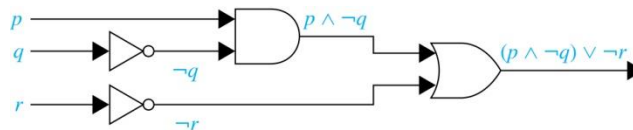
Logic Circuits

(Studied in depth in Chapter 12)

- Electronic circuits; each input/output signal can be viewed as a 0 or 1.
 - 0 represents **False**
 - 1 represents **True**
- Complicated circuits are constructed from three basic circuits called gates.

- 
 - The **inverter** takes one input bit and produces the value equivalent to the negation of that bit.
- 
 - The **OR gate** takes two input bits and produces the value equivalent to the disjunction of the two bits.
- 
 - The **AND gate** takes two input bits and produces the value equivalent to the conjunction of the two bits.

- More complicated digital circuits can be constructed by combining these basic circuits to produce the desired output given the input signals by building a circuit for each piece of the output expression and then combining them. For example:



Propositional Equivalences

Section 1.3

Tautologies(恆真) , Contradictions(矛盾), and Contingencies(偶發)

- A **tautology** is a proposition which is **always true**.
 - Example: $p \vee \neg p$
- A **contradiction** is a proposition which is **always false**.
 - Example: $p \wedge \neg p$
- A **contingency** is a proposition which is **neither a tautology nor a contradiction**, such as $\neg p$

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logically Equivalent

- Two compound propositions **p and q are logically equivalent** if **$p \leftrightarrow q$ is a tautology**.
- We write this as $p \leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws



De Morgan's 1ST Laws $\neg(p \wedge q) \equiv \neg p \vee \neg q$

De Morgan's 2ND Laws $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Augustus De Morgan

1806-1871

This truth table shows that De Morgan's 2ND Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences

- Identity Laws: $p \wedge T \equiv p$, $p \vee F \equiv p$
(同一律)
- Domination Laws: $p \vee T \equiv T$, $p \wedge F \equiv F$
(支配律)
- Idempotent laws: $p \vee p \equiv p$, $p \wedge p \equiv p$
(幂等律)
- Double Negation Law: $\neg(\neg p) \equiv p$
(雙重否定律)
- Negation Laws: $p \vee \neg p \equiv T$, $p \wedge \neg p \equiv F$
(否定律)

Key Logical Equivalences (*cont*)

- Commutative Laws: $p \vee q \equiv q \vee p$
(交換律) $p \wedge q \equiv q \wedge p$
- Associative Laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
(結合律) $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- Distributive Laws: $(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$
(分配律) $(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$
- Absorption Laws: $p \vee (p \wedge q) \equiv p$
(吸收律) $p \wedge (p \vee q) \equiv p$

More Logical Equivalences

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Constructing New Logical Equivalences

- We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.
- To prove that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B.

$$\begin{array}{c} A \equiv A_1 \\ \vdots \\ A_n \equiv B \end{array}$$

Equivalence Proofs

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q))$	$\equiv$	$\neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv$	$\neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv$	$\neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv$	$(\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv$	$F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
	$\equiv$	$(\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
	$\equiv$	$(\neg p \wedge \neg q)$	by the identity law for F

Equivalence Proofs

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology (恆真).

Solution:

$(p \wedge q) \rightarrow (p \vee q)$	$\equiv$	$\neg(p \wedge q) \vee (p \vee q)$	by truth table for $\rightarrow$
	$\equiv$	$(\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
	$\equiv$	$(\neg p \vee p) \vee (q \vee \neg q)$	by associative and commutative laws
			laws for disjunction
	$\equiv$	$T \vee T$	by truth tables
	$\equiv$	T	by the domination law

The Foundations: Logic and Proofs

Chapter 1, Part II: Predicate Logic (謂詞邏輯/述詞邏輯)

Edited by ShihTsung Liang

Predicates(謂詞/述詞) and Quantifiers(量詞)

Section 1.4

Propositional Logic Not Enough

- If we have:
 - “All men are mortal.”
 - “Socrates is a man.”
- Does it follow that “Socrates is mortal?”
- Can’t be represented in propositional logic.
Need a language that talks about objects,
their properties, and their relations.
- Later we’ll see how to draw inferences.

Introducing Predicate Logic

- The atoms of **propositional logic** are simple statements.
- The atoms of **predicate logic** are statements involving variables (which can be **quantified**) and predicates
 - e.g., **x is greater than zero.**
- An atomic statement is denoted by a **propositional function**
 - e.g., $P(x)$: x is greater than zero.

Propositional Functions

- **Propositional functions** become propositions (and have truth values) when their variables are each **replaced by a value** from the *domain* (or ***bound by a quantifier***, as we will see later).
- For example, let $P(x)$ denote “ $x > 0$ ” and the domain be the integers. Then:
 - $P(-3)$ is false.
 - $P(0)$ is false.
 - $P(3)$ is true.
- Often the **domain** is denoted by U . So in this example U is the integers.

Examples of Propositional Functions

- Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

$R(2, -1, 5)$

Solution: F

$R(3, 4, 7)$

Solution: T

$R(x, 3, z)$

Solution: Not a Proposition

- Now let “ $x - y = z$ ” be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

$Q(2, -1, 3)$

Solution: T

$Q(3, 4, 7)$

Solution: F

$Q(x, 3, z)$

Solution: Not a Proposition

Compound Expressions

- **Connectives** from propositional logic carry over to predicate logic.
- If $P(x)$ denotes “ $x > 0$,” find these truth values:
 - $P(3) \vee P(-1)$ **Solution:** T
 - $P(3) \wedge P(-1)$ **Solution:** F
 - $P(3) \rightarrow P(-1)$ **Solution:** F
 - $P(-3) \rightarrow P(-1)$ **Solution:** T
- Expressions **with variables** are **not propositions** and therefore do not have truth values. For example,
 - $P(3) \wedge P(y)$
 - $P(x) \rightarrow P(y)$
- When used with **quantifiers** (to be introduced next), these expressions (propositional functions) become propositions.

Quantifiers



Charles Peirce (1839-1914)

- We need *quantifiers* to express the meaning of English words including *all* and *some*:
 - “All men are Mortal.”
 - “Some cats do not have fur.”
- The two most important quantifiers are:
 - *Universal Quantifier*(全称量詞), “For all,” symbol: $\forall$
 - *Existential Quantifier*(存在量詞), “There exists,” symbol: $\exists$
- $\forall x P(x)$ asserts $P(x)$ is true for every x in the *domain*.
- $\exists x P(x)$ asserts $P(x)$ is true for some x in the *domain*.
- The *quantifiers* are said to *bind the variable x* in these expressions.

Universal Quantifier (全称量词)

- $\forall x P(x)$ is read as “For all x , $P(x)$ ” or “For every x , $P(x)$ ”

Examples:

- 1) If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\forall x P(x)$ is false.
- 2) If $P(x)$ denotes “ $x > 0$ ” and U is the positive integers, then $\forall x P(x)$ is true.
- 3) If $P(x)$ denotes “ x is even” and U is the integers, then $\forall x P(x)$ is false.

Existential Quantifier (存在量詞)

- $\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

Properties of Quantifiers

- The truth value of $\exists x P(x)$ and $\forall x P(x)$ depends both on the propositional function $P(x)$ and on the domain U .
- **Examples:**
 1. If U is the positive integers and $P(x)$ is the statement “ $x < 2$ ”, then $\exists x P(x)$ is true, but $\forall x P(x)$ is false.
 2. If U is the negative integers and $P(x)$ is the statement “ $x < 2$ ”, then both $\exists x P(x)$ and $\forall x P(x)$ are true.
 3. If U consists of 3, 4, and 5, and $P(x)$ is the statement “ $x > 2$ ”, then both $\exists x P(x)$ and $\forall x P(x)$ are true. But if $P(x)$ is the statement “ $x < 2$ ”, then both $\exists x P(x)$ and $\forall x P(x)$ are false.

Precedence(優先順序) of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.
- For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$
- $\forall x (P(x) \vee Q(x))$ means something different.
- Unfortunately, often people write $\forall x P(x) \vee Q(x)$ when they mean $\forall x (P(x) \vee Q(x))$.

Equivalences in Predicate Logic

- Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value
 - for **every predicate substituted into these statements** and
 - for **every domain of discourse** used for the *variables* in the expressions.
- The notation $S \equiv T$ indicates that S and T are logically equivalent.
- **Example:** $\forall x \neg \neg S(x) \equiv \forall x S(x)$

Thinking about Quantifiers as Conjunctions(合取) and Disjunctions(分取)

- If the **domain is finite**, a **universally quantified proposition** is equivalent to a **conjunction** of propositions without quantifiers and an **existentially quantified proposition** is equivalent to a **disjunction** of propositions without quantifiers.
- If U consists of the integers 1, 2, and 3:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

- Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

Negating(否定) Quantified Expressions

- Consider $\forall x J(x)$
“Every student in your class has taken a course in calculus.”
Here $J(x)$ is “x has taken a course in calculus” and
the domain is students in your class.
- Negating the original statement gives “It is not the case that every student in your class has taken calculus.” This implies that “There is a student in your class who has not taken calculus.”
Symbolically $\neg \forall x J(x)$ and $\exists x \neg J(x)$ are equivalent

Negating Quantified Expressions (continued)

- Now Consider $\exists x J(x)$
“There is a student in this class who has taken a course in Java.”
Where $J(x)$ is “x has taken a course in Java.”
- Negating the original statement gives “It is not the case that there is a student in this class who has taken Java.” This implies that “Every student in this class has not taken Java”
Symbolically $\neg \exists x J(x)$ and $\forall x \neg J(x)$ are equivalent

De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are:

TABLE 2 De Morgan's Laws for Quantifiers.			
<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

- Summary:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

Translation from English to Logic

Examples:

1. “Some student in this class has visited Mexico.”

Solution: Let $M(x)$ denote “ x has visited Mexico” and $S(x)$ denote “ x is a student in this class,” and U be all people.

$$\exists x (S(x) \wedge M(x))$$

****note:** $\exists x (S(x) \rightarrow M(x))$ is incorrect. What does it mean?

2. “Every student in this class has visited Canada or Mexico.”

Solution: Add $C(x)$ denoting “ x has visited Canada.”

$$\forall x (S(x) \rightarrow (M(x) \vee C(x)))$$

****note:** $\forall x (S(x) \wedge (M(x) \vee C(x)))$ is incorrect!!! What does it mean?

System Specification Example

(Reading Exercise)

- Predicate logic is used for specifying properties that systems must satisfy.
- For example, translate into predicate logic:
 - “Every mail message larger than one megabyte will be compressed.”
 - “If a user is active, at least one network link will be available.”
- Decide on predicates and domains (left implicit here) for the variables:
 - Let $L(m, y)$ be “Mail message m is larger than y megabytes.”
 - Let $C(m)$ denote “Mail message m will be compressed.”
 - Let $A(u)$ represent “User u is active.”
 - Let $S(n, x)$ represent “Network link n is state x .”
- Now we have:

$$\forall m (L(m, 1) \rightarrow C(m))$$
$$\exists u A(u) \rightarrow \exists n S(n, available)$$

Lewis Carroll Example



Lewis Carroll
(1832-1898)

- The first two are called *premises* and the third is called the *conclusion*.
 1. “All lions are fierce.”
 2. “Some lions do not drink coffee.”
 3. “Some fierce creatures do not drink coffee.”
- Here is one way to translate these statements to predicate logic. Let $P(x)$, $Q(x)$, and $R(x)$ be the propositional functions “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively.
 1. $\forall x (P(x) \rightarrow Q(x))$
 2. $\exists x (P(x) \wedge \neg R(x))$
 3. $\exists x (Q(x) \wedge \neg R(x))$
- Later we will see how to prove that the conclusion follows from the premises (*rule of inferences*)

Nested Quantifiers

Section 1.5

Nested Quantifiers

- Nested quantifiers are often necessary to express the meaning of sentences in English as well as important concepts in computer science and mathematics.

Example: “Every real number has an inverse” is

$$\forall x \exists y (x + y = 0)$$

where the domains of x and y are the real numbers.

- We can also think of nested propositional functions:
 $\forall x \exists y (x + y = 0)$ can be viewed as $\forall x Q(x)$ where $Q(x)$ is $\exists y P(x, y)$ where $P(x, y)$ is $(x + y = 0)$

Order of Quantifiers

Examples:

1. Let $P(x,y)$ be the statement “ $x + y = y + x$.”
Assume that U is the real numbers. Then $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$ have the same truth value.
2. Let $Q(x,y)$ be the statement “ $x + y = 0$.”
Assume that U is the real numbers. Then $\forall x \exists y Q(x,y)$ is true, but $\exists y \forall x Q(x,y)$ is false.

Questions on Order of Quantifiers

Example 1: Let U be the real numbers,

Define $P(x,y) : x \cdot y = 0$

What is the truth value of the following:

1. $\forall x \forall y P(x,y)$

Answer: False

2. $\forall x \exists y P(x,y)$

Answer: True

3. $\exists x \forall y P(x,y)$

Answer: True

4. $\exists x \exists y P(x,y)$

Answer: True

Questions on Order of Quantifiers

Example 2: Let U be the real numbers,

Define $P(x,y) : x / y = 1$

What is the truth value of the following:

1. $\forall x \forall y P(x,y)$

Answer: False

2. $\forall x \exists y P(x,y)$

Answer: False

3. $\exists x \forall y P(x,y)$

Answer: False

4. $\exists x \exists y P(x,y)$

Answer: True

Quantifications of Two Variables

(Reading Exercise)

Statement	When True?	When False
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y

Translating Nested Quantifiers into English

Example 1: Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

where $C(x)$ is “ x has a computer,” and $F(x,y)$ is “ x and y are friends,” and the domain for both x and y consists of all students in your school.

Solution: Every student in your school has a computer or has a friend who has a computer.

Example 1: Translate the statement

$$\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

Solution: There is a student none of whose friends are also friends with each other.

Translating Mathematical Statements into Predicate Logic (Reading Exercise)

Example : Translate “The sum of two positive integers is always positive” into a logical expression.

Solution:

1. Rewrite the statement to make the implied quantifiers and domains explicit:

“For every two integers, if these integers are both positive, then the sum of these integers is positive.”

2. Introduce the variables x and y , and specify the domain, to obtain:

“For all positive integers x and y , $x + y$ is positive.”

3. The result is:

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0))$$

where the domain of both variables consists of all integers

Translating English into Logical Expressions Example

Example: Use quantifiers to express the statement
“There is a woman who has taken a flight (航班)
on every airline(航空公司) in the world.”

Solution:

1. Let $P(w,f)$ be “ w has taken f ” and $Q(f,a)$ be “ f is a flight on a .”
2. The domain of w is all women, the domain of f is all flights, and the domain of a is all airlines.
3. Then the statement can be expressed as:

$$\exists w \forall a \exists f (P(w,f) \wedge Q(f,a))$$

Negating Nested Quantifiers

Example 1: Recall the logical expression developed in the previous slide:

$$\exists w \forall a \exists f (P(w,f) \wedge Q(f,a))$$

Part 1: Use quantifiers to express the statement that “There does not exist a woman who has taken a flight on every airline in the world.”

Solution: $\neg \exists w \forall a \exists f (P(w,f) \wedge Q(f,a))$

Part 2: Now use De Morgan’s Laws to move the negation as far inwards as possible.

Solution:

1. $\neg \exists w \forall a \exists f (P(w,f) \wedge Q(f,a))$
2. $\forall w \neg \forall a \exists f (P(w,f) \wedge Q(f,a))$ by De Morgan’s for $\exists$
3. $\forall w \exists a \neg \exists f (P(w,f) \wedge Q(f,a))$ by De Morgan’s for $\forall$
4. $\forall w \exists a \forall f \neg (P(w,f) \wedge Q(f,a))$ by De Morgan’s for $\exists$
5. $\forall w \exists a \forall f (\neg P(w,f) \vee \neg Q(f,a))$ by De Morgan’s for $\wedge$.

Part 3: Can you translate the result back into English?

Solution:

“For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline”

The Foundations: Logic and Proofs

Chapter 1, Part III: Proofs

Edited by Shih-Tsung Liang

Rules of Inference

Section 1.6

Revisiting the Socrates Example

- We have the two premises(前提):
 - “All men are mortal.”
 - “Socrates is a man.”
- And the conclusion(結論):
 - “Socrates is mortal.”
- How do we get the conclusion from the premises?

The Argument (論證)

- We can express the premises (above the line) and the conclusion (below the line) in predicate logic as an argument:

$$\forall x (Man(x) \rightarrow Mortal(x))$$

$$Man(Socrates)$$

$$\therefore Mortal(Socrates)$$

- We will see shortly that this is a valid(有效的) argument.

Arguments in Propositional Logic

- An *argument* in propositional logic is a sequence of propositions. All but the final proposition are called *premises*. The final proposition is the *conclusion*.
- An argument is valid if the truth of all its premises imply that the conclusion is true.
- The argument with premises $p_1, p_2, \dots, p_n$ and the conclusion q is valid if
$$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q \text{ is a tautology.}$$

Rules of Inference for Propositional Logic: Modus Ponens (離斷律)

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

Example:

Let p be “It is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“It is snowing.”

“Therefore , I will study discrete math.”

Modus Tollens (逆斷律)

$$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$$

Corresponding Tautology:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

Example:

Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore, it is not snowing.”

Hypothetical Syllogism (假設三段論)

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$$

Corresponding Tautology:
 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Example:

Let p be “it snows.”

Let q be “I will study discrete math.”

Let r be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore , If it snows, I will get an A.”

Disjunctive Syllogism(分取三段論)

$$\frac{p \vee q \quad \neg p}{\therefore q}$$

Corresponding Tautology:

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math or I will study English literature.”

“I will not study discrete math.”

“Therefore , I will study English literature.”

Addition (添加律)

$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math or I will visit Las Vegas.”

Simplification (簡化律)

$$\frac{p \wedge q}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math and English literature”

“Therefore, I will study discrete math.”

Conjunction (合取律)

$$\frac{p}{q} \quad \frac{q}{\hline \therefore p \wedge q}$$

Corresponding Tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math.”

“I will study English literature.”

“Therefore, I will study discrete math and I will study English literature.”

Resolution (預解律)

Resolution plays an important role in AI and is used in Prolog.

$$\frac{\neg p \vee r \quad p \vee q}{\therefore q \vee r}$$

Corresponding Tautology:

$$((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$$

Example:

Let p be “I will study discrete math.”

Let r be “I will study English literature.”

Let q be “I will study databases.”

“I will not study discrete math or I will study English literature.”

“I will study discrete math or I will study databases.”

“Therefore, I will study databases or I will English literature.”

Using the Rules of Inference to Build Valid Arguments (有效論證)

- A *valid argument* is a sequence of statements. Each statement is either a **premise** or follows from previous statements by **rules of inference**. The last statement is called **conclusion**.

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Valid Arguments

Example 2:

- With these hypotheses:
 - “It is not sunny this afternoon and it is colder than yesterday.”
 - “We will go swimming only if it is sunny.”
 - “If we do not go swimming, then we will take a canoe (獨木舟) trip.”
 - “If we take a canoe trip, then we will be home by sunset.”
- Using the inference rules, construct a valid argument for the conclusion:
 - “We will be home by sunset.”

Solution:

1. Choose propositional variables:

p : “It is sunny this afternoon.”

r : “We will go swimming.”

t : “We will be home by sunset.”

q : “It is colder than yesterday.”

s : “We will take a canoe trip.”

2. Translation into propositional logic:

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

Continued on next slide →

Valid Arguments

Hypotheses: $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, $s \rightarrow t$

Conclusion: t

3. Construct the Valid Argument

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Handling Quantified Statements

- Rules of Inference for Quantified Statements

TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation (全稱個例化;UI)
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$	Universal generalization (全稱通則化;UG)
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation (存在個例化;EI)
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential generalization (存在通則化;EG)

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

“Therefore, Fido is cuddly.”

Used often implicitly in Mathematical Proofs.

“Someone got an A in the course.”

“Let’s call her *a* and say that *a* got an A”

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

Using Rules of Inference

Example 1: Using the rules of inference, construct a valid argument to show that

“John Smith has two legs”

is a consequence of the premises:

“Every man has two legs.” “John Smith is a man.”

Solution: Let $M(x)$ denote “ x is a man” and

$L(x)$ “ x has two legs” and

John Smith be a member of the domain.

Valid Argument:

Step	Reason
1. $\forall x(M(x) \rightarrow L(x))$	Premise
2. $M(J) \rightarrow L(J)$	UI from (1)
3. $M(J)$	Premise
4. $L(J)$	Modus Ponens using (2) and (3)

Using Rules of Inference

(Reading Exercise)

Example 2: Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

Solution: Let $C(x)$ denote “ x is in this class,”

$B(x)$ denote “ x has read the book,” and

$P(x)$ denote “ x passed the first exam.”

First we translate the premises and conclusion into symbolic form.

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Continued on next slide →

Using Rules of Inference

(Reading Exercise)

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Construct the Valid Argument:

Step	Reason
1. $\exists x(C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	EI from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x(C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	UI from (4)
6. $P(a)$	MP from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conj from (6) and (7)
9. $\exists x(P(x) \wedge \neg B(x))$	EG from (8)

Returning to the Socrates Example

$$\forall x(Man(x) \rightarrow Mortal(x))$$

$$Man(Socrates)$$

$$\therefore Mortal(Socrates)$$

Solution for Socrates Example

Valid Argument

Step

1. $\forall x(Man(x) \rightarrow Mortal(x))$

2. $Man(Socrates) \rightarrow Mortal(Socrates)$

3. $Man(Socrates)$

4. $Mortal(Socrates)$

Reason

Premise

UI from (1)

Premise

MP from (2)
and (3)

Universal Modus Ponens (全稱離斷律)

Universal Modus Ponens combines universal instantiation and modus ponens into one rule.

$$\frac{\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular} \\ \text{element in the domain} \end{array}}{\therefore Q(a)}$$

This rule could be used in the Socrates example.

Introduction to Proofs

Section 1.7

Proofs of Mathematical Statements

- A *proof* is a **valid argument** (有效論證) that establishes the **truth of a statement**.
- In math, CS, and other disciplines, **informal proofs** which are generally shorter, are generally used.
 - **More than one rule** of inference are often used **in a step**.
 - **Steps** may be **skipped**.
 - The **rules** of inference used are **not explicitly stated**.
 - for to understand and to explain **to people** (not **computer**).
 - But it is also **easier to introduce errors**.

Definitions

- A *theorem* (定理) is a statement that can be shown to be true using:
 - definitions
 - other theorems
 - *Axioms* (公理) (statements which are *given as true*)
 - rules of inference
- A *lemma* (輔助定理) is a ‘*helping theorem*’ or a result which *is needed to prove a theorem*.
- A *corollary* (推論) is a result which *follows directly from a theorem*.
- A *conjecture* (猜想) is a statement that is being proposed to be true. Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false.

How Theorems are Stated

- Often the **universal quantifier** (needed for a precise statement of a theorem) **is omitted** by standard mathematical convention.

For example, the statement:

“If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$ ”

really means

“**For all** positive real numbers x and y , if $x > y$, then $x^2 > y^2$.”

Methods of Proving Theorems

- Many theorems have the form:

$$\forall x(P(x) \rightarrow Q(x))$$

- To prove them, we show that where c is an arbitrary element of the domain, $P(c) \rightarrow Q(c)$
- By **universal generalization** the truth of the original formula follows.
- So, we must prove something of the form: $p \rightarrow q$

Proving Conditional Statements: $p \rightarrow q$

- **Direct Proof**: Assume that p is true. Use rules of inference, axioms, and logical equivalences to show that q must also be true.

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Solution: Assume that n is odd. Then $n = 2k + 1$ for an integer k . Squaring both sides of the equation, we get:
$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1,$$
where $r = 2k^2 + 2k$, an integer.

We have proved that if n is an odd integer, then n^2 is an odd integer. ◀

(◀ marks the end of the proof. Sometimes **QED** is used instead.)

Proving Conditional Statements: $p \rightarrow q$

Definition: The **real number** r is **rational** if there exist integers p and q where $q \neq 0$ such that $r = p/q$

Example: Prove that the sum of two rational numbers is rational.

Solution: Assume r and s are two rational numbers.

Then there must be integers p, q and also t, u such

that $r = p/q, s = t/u, u \neq 0, q \neq 0$

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu} = \frac{v}{w} \quad \text{where } v = pu + qt \\ w = qu \neq 0$$

Thus the sum is rational.



Proving Conditional Statements: $p \rightarrow q$

- Proof by Contraposition(對位): Assume $\neg q$ and show $\neg p$ is true also. This is sometimes called an *indirect proof* method. If we give a direct proof of $\neg q \rightarrow \neg p$ then we have a proof of $p \rightarrow q$.

Why does this work? (Recall text p.8, or Lecture p.14)

Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution: Assume n is even. So, $n = 2k$ for some integer k . Thus

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j \text{ for } j = 3k + 1$$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even), then n is odd (not even). ◀

Proving Conditional Statements: $p \rightarrow q$

(Reading Exercise)

Example: Prove that for an integer n , if n^2 is odd, then n is odd.

Solution: Use proof by contraposition. Assume n is even (i.e., not odd). Therefore, there exists an integer k such that $n = 2k$. Hence,

$$n^2 = 4k^2 = 2(2k^2)$$

and n^2 is even (i.e., not odd).

We have shown that if n is an even integer, then n^2 is even. Therefore by contraposition, for an integer n , if n^2 is odd, then n is odd.



Proof by Contradiction

- **Example:** Use a proof by contradiction to give a proof that $\sqrt{2}$ is irrational.

Solution: Suppose $\sqrt{2}$ is rational. Then there exists integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and **a and b have no common factors** Then

$$2 = \frac{a^2}{b^2} \qquad 2b^2 = a^2$$

Therefore a^2 must be even. If a^2 is even then a must be even (an exercise). Since a is even, $a = 2c$ for some integer c . Thus,

$$2b^2 = 4c^2 \qquad b^2 = 2c^2$$

Therefore b^2 is even. Again then b must be even as well.

But then 2 must divide both a and b . This contradicts our assumption that a and b have no common factors.

We have proved by contradiction that our initial assumption must be false and therefore $\sqrt{2}$ is irrational.



Proof by Contradiction

(Reading Exercise)

- Proof by Contradiction(矛盾):

To prove p , assume $\neg p$ and derive a contradiction such as $r \wedge \neg r$ (an indirect form of proof). Since we have shown that $\neg p \rightarrow \mathbf{F}$ is true, it follows that the contrapositive(對位命題) $\mathbf{T} \rightarrow p$ also holds.

Example: Show that at least 4 of any 22 days must fall on the same day of the week.

Solution: Let p be the proposition “at least 4 of any 22 days must fall on the same day of the week” and r be the proposition “22 days are chosen”

Suppose that $\neg p$ is True. This means that no more than 3 of the 22 days fall on the same day of the week. Because there are 7 days of the week, we could only have picked 21 days. This contradicts the assumption that we have picked 22 days.



Theorems that are Biconditional Statements

- To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.

Example: Prove the theorem: “If n is an integer, then n is odd **if and only** if n^2 is odd.”

Solution: We have already shown (Lecture P98 & P101) that both $p \rightarrow q$ and $q \rightarrow p$. Therefore we can conclude $p \leftrightarrow q$.

Sometimes **iff** is used as an abbreviation for “**if an only if**,” as in
“If n is an integer, then n is odd iff n^2 is odd.”

Looking Ahead

- If direct methods of proof do not work:
 - We may need a clever use of a proof by **contraposition**.
 - Or a proof by **contradiction**.
 - In the next section, we will see **strategies** that can be used when straightforward approaches do not work.
 - In Chapter 5, we will see **mathematical induction** and related techniques.
 - In Chapter 6, we will see **combinatorial proofs**

Proof Methods and Strategy

Section 1.8

Proof by Cases

- To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

- Use the tautology

$$\begin{aligned} [(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] &\leftrightarrow \\ &[(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)] \end{aligned}$$

- Each of the implications $p_i \rightarrow q$ is a *case*.

Proof by Cases

Example: Let $a @ b = \max\{a, b\} = a$ if $a \geq b$, otherwise $a @ b = \max\{a, b\} = b$.
Show that for all real numbers a, b, c

$$(a @ b) @ c = a @ (b @ c)$$

(This means the operation $@$ is associative.)

Proof: Let a, b , and c be arbitrary real numbers.
Then one of the following 6 cases must hold.

1. $a \geq b \geq c$
2. $a \geq c \geq b$
3. $b \geq a \geq c$
4. $b \geq c \geq a$
5. $c \geq a \geq b$
6. $c \geq b \geq a$

Continued on next slide →

Proof by Cases

Case 1: $a \geq b \geq c$

$(a @ b) = a, a @ c = a, b @ c = b$

Hence $(a @ b) @ c = a = a @ (b @ c)$

Therefore the equality holds for the first case.

A complete proof requires that the equality be shown to hold for all 6 cases. But the proofs of the remaining cases are similar. Try them.



Without Loss of Generality

Example: Show that if x and y are integers and both $x \cdot y$ and $x + y$ are even, then both x and y are even.

Proof: Use a proof by contraposition(對位). Suppose x and y are not both even. Then, one or both are odd. **Without loss of generality**, assume that x is odd. Then $x = 2m + 1$ for some integer k .

Case 1: y is even. Then $y = 2n$ for some integer n , so $x + y = (2m + 1) + 2n = 2(m + n) + 1$ is odd.

Case 2: y is odd. Then $y = 2n + 1$ for some integer n , so $x \cdot y = (2m + 1)(2n + 1) = 2(2m \cdot n + m + n) + 1$ is odd.

We only cover the case where x is odd because the case where y is odd is similar. The use phrase *without loss of generality* (WLOG) indicates this.

Existence Proofs



Srinivasa Ramanujan
(1887-1920)

- Proof of theorems of the form $\exists x P(x)$.
- **Constructive existence proof:**
 - Find an explicit value of c , for which $P(c)$ is true.
 - Then $\exists x P(x)$ is true by Existential Generalization (EG).

Example: Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways:

Proof: 1729 is such a number since
$$1729 = 10^3 + 9^3 = 12^3 + 1^3$$



Godfrey Harold Hardy
(1877-1947)

Nonconstructive Existence Proofs

- In a *nonconstructive existence proof*, we **assume** no c exists which makes $P(c)$ true and derive a contradiction.

Example: Show that there exist irrational numbers x and y such that x^y is rational.

Proof: We know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$.

If it is rational, we have two irrational numbers x and y with x^y rational, namely $x = \sqrt{2}$ and $y = \sqrt{2}$.

But if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2} \sqrt{2})} = \sqrt{2}^2 = 2$.



Uniqueness Proofs

- Some theorems assert the existence of a unique element with a particular property, $\exists! x P(x)$. The two parts of a *uniqueness proof* are
 - **Existence**: We show that an element x with the property exists.
 - **Uniqueness**: We show that if $y \neq x$, then y does not have the property.

Example: Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Solution:

- Existence: The real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a) + b = -b + b = 0$.
- Uniqueness: Suppose that s is a real number such that $as + b = 0$. Then $ar + b = as + b$, where $r = -b/a$. Subtracting b from both sides and dividing by a shows that $r = s$.



Counterexamples (反例)

- Recall $\exists x \neg P(x) \equiv \neg \forall x P(x)$.
- To establish that $\forall x P(x)$ is false (or $\neg \forall x P(x)$ is true) find a c such that $\neg P(c)$ is true (or $P(c)$ is false).
- In this case c is called a *counterexample* to the assertion $\forall x P(x)$.

Example: “Every positive integer is the sum of the squares of 3 integers.” The integer 7 is a counterexample. So the claim is false.

Proof Strategies for proving $p \rightarrow q$

- Choose a method.
 1. First try a **direct method of proof**.
 2. If this does not work, try an **indirect method** (e.g., try to prove the contrapositive).
- For whichever method you are trying, choose a strategy.
 1. First try **forward reasoning**. Start with the axioms and known theorems and construct a sequence of steps that end in the conclusion. Start with p and prove q , or start with $\neg q$ and prove $\neg p$.
 2. If this doesn't work, try **backward reasoning**. When trying to prove q , find a statement p that we can prove with the property $p \rightarrow q$.

Backward Reasoning

Example: Suppose that two people play a game **taking turns removing, 1, 2, or 3 stones** at a time from a pile that **begins with 15 stones**. The person who **removes the last stone wins the game**. Show that the first player can win the game no matter what the second player does.

Proof: Let n be the last step of the game.

Step n : Player₁ can win if the pile contains 1, 2, or 3 stones.

Step $n-1$: Player₂ will have to leave such a pile if the pile that he/she is faced with has 4 stones.

Step $n-2$: Player₁ can leave 4 stones when there are 5, 6, or 7 stones left at the beginning of his/her turn.

Step $n-3$: Player₂ must leave such a pile, if there are 8 stones.

Step $n-4$: Player₁ has to have a pile with 9, 10, or 11 stones to ensure that there are 8 left.

Step $n-5$: Player₂ needs to be faced with 12 stones to be forced to leave 9, 10, or 11.

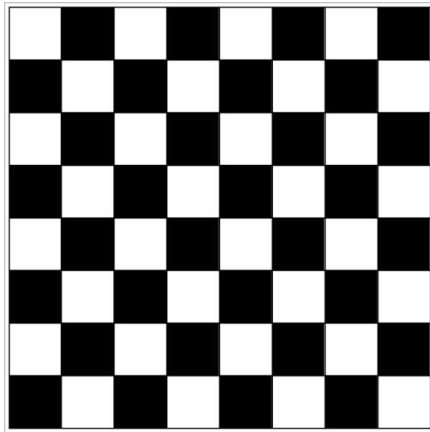
Step $n-6$: Player₁ can leave 12 stones by removing 3 stones.

Now reasoning forward, the first player can ensure a win by removing 3 stones and leaving 12.

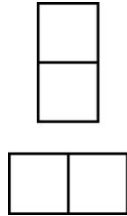
Proof and Disproof: Tilings

Example 1: Can we tile the standard checkerboard using dominos?

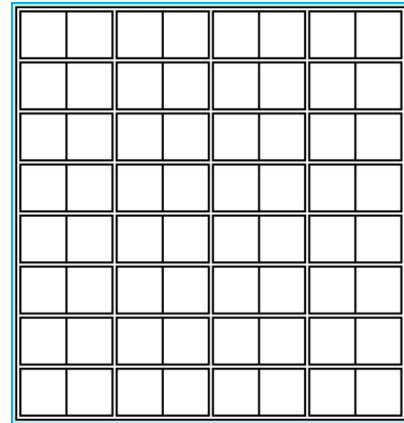
Solution: Yes! One example provides a constructive existence proof.



The Standard Checkerboard



Two Dominoes



One Possible Solution

Tilings

Example 2: Can we tile a checkerboard obtained by removing one of the four corner squares of a standard checkerboard?

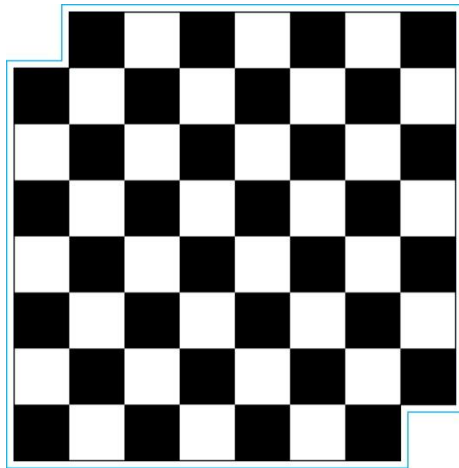
Solution:

- Our checkerboard has $64 - 1 = 63$ squares.
- Since each domino has two squares, a board with a tiling must have an even number of squares.
- The number 63 is not even.
- We have a contradiction.

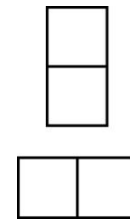


Tilings

Example 3: Can we tile a board obtained by removing both the upper left and the lower right squares of a standard checkerboard?



Nonstandard Checkerboard



Dominoes

Continued on next slide →

Tilings

Solution:

- There are 62 squares in this board.
- To tile it we need 31 dominos.
- *Key fact:* Each domino covers one black and one white square.
- Therefore the tiling covers 31 black squares and 31 white squares.
- Our board has either 30 black squares and 32 white squares or 32 black squares and 30 white squares.
- Contradiction!



The Role of Open Problems

- Unsolved problems have motivated much work in mathematics. **Fermat's Last Theorem** was conjectured more than 300 years ago. It has only recently been finally solved.

Fermat's Last Theorem: The equation $x^n + y^n = z^n$ has no solutions in integers x , y , and z , with $xyz \neq 0$ whenever n is an integer with $n > 2$.

A proof was found by Andrew Wiles in the 1990s.

Ex. $x \in \mathbb{Z}^+$, if $1285^x - 1116^x = 637^x$. Find x .

An Open Problem

- **The $3x + 1$ Conjecture:** Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x + 1$. For all positive integers x , when we repeatedly apply the transformation T , we will eventually reach the integer 1.

For example, starting with $x = 13$:

$$T(13) = 3 \cdot 13 + 1 = 40, T(40) = 40/2 = 20, T(20) = 20/2 = 10,$$

$$T(10) = 10/2 = 5, T(5) = 3 \cdot 5 + 1 = 16, T(16) = 16/2 = 8,$$

$$T(8) = 8/2 = 4, T(4) = 4/2 = 2, T(2) = 2/2 = 1$$

The conjecture has been verified using computers up to $5.6 \cdot 10^{13}$.

Additional Proof Methods

- Later we will see many other proof methods:
 - **Mathematical induction (數學歸納法)**, which is a useful method for proving statements of the form $\forall n P(n)$, where the domain consists of all **positive integers**.
 - **Structural induction (結構歸納法)**, which can be used to prove such results about recursively defined **sets**.
 - **Cantor diagonalization (Cantor對角法)** is used to prove results about the size of infinite sets.
 - **Combinatorial proofs (組合證明法)** use counting arguments.